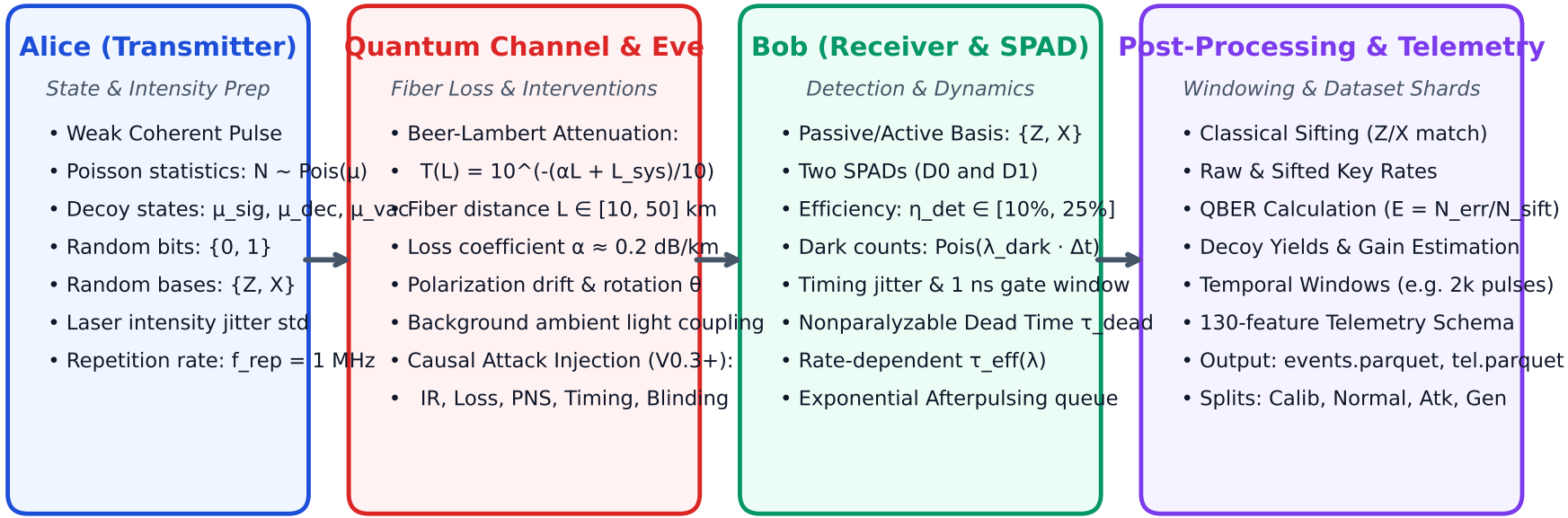
QISM Simulation Pipeline: Physics-Informed Stochastic QKD Telemetry

Decoy-State BB84 Protocol with Fiber Channel, SPAD Dynamics, Physical Attack Injection, and Sharded Telemetry



Scientific Invariant: Causal Propagation

Attack and environmental drift causally perturb physical simulation variables (photons, transmittances, efficiencies) -> events -> windowed telemetry. Telemetry features are never modified directly.